

Assignment – Unit 1

Subject: Programming for Problem Solving

Subject Code: 25CSH-107

Semester: 1

Group Distribution by UID:

Group	Assigned UIDs
Group A	First 1 – 7 UID's
Group B	Next 8 – 13 UID's
Group C	Next 14 – 19 UID's
Group D	Next 20 – 25 UID's
Group E	Next 26 – 30 UID's
Group F	Next 31 – 34 UID's
Group G	Next 35 – 37 UID's

Group A:

1. Design a flowchart and write a program to check if a number is a Duck number, ignoring leading zeros.
2. Develop a C program to input and validate integer, float, character, and string types with appropriate prompts and error messages.
3. Implement a program using nested ternary operators to find the largest of four numbers.
4. Create a program to input a float, cast it to integer, and perform mathematical operations on both types.
5. Write a program using logical conditions to classify a number as prime, composite, or invalid (for <2).

Group B:

1. Create flowchart components like loops, conditions, sub-processes with their roles using a diagram with the help of example.
2. Write a C program to compute weighted average of marks based on subject credits and print performance classification.
3. Input a string and print ASCII values of each character using a loop.
4. Write an expression evaluator in C that inputs operators and operands, and prints result with precedence handling.
5. Create a program using switch-case to simulate a basic ATM menu (check balance, deposit, withdraw, exit).

Group C:

1. Write an algorithm to generate Fibonacci series using a flowchart and implement it using a loop in C.
2. Create a C program that converts any input uppercase word to lowercase and checks for palindrome.
3. Input age and income; use logical conditions to check loan eligibility (age > 21 && income > 25000).
4. Design a format printer that aligns output in columns using \t and \% formatting.
5. Implement a grading system using if-else-if that also validates score input (0–100).

Group D:

1. Design a flowchart and C program to compute average speed using distance and time input.
2. Write a C program to display data type sizes using macros and `sizeof()` operator.
3. Develop a truth table generator using logical and relational operators for two input variables.
4. Convert a given number of hours into days, weeks, and years considering leap years.
5. Write a character classifier using nested-if conditions and print category based on extended ASCII ranges.

Group E:

1. Write a structured algorithm and program to determine the type of triangle (Equilateral, Isosceles, Scalene).
2. Implement a fine calculator for a digital library using enhanced if-else or switch-case structures.

3. Write a function to determine whether a number is perfect, abundant, or deficient.
4. Debug and explain output of complex conditional statements with nested ternary operators.
5. Build a basic compiler simulation that reads comma expressions and evaluates rightmost assignment.

Group F:

1. Create a flowchart to determine whether a 4-digit number is an Armstrong number.
2. Write a formatted table display using nested loops and escape sequences (multiplication table).
3. Develop a digit separator function that prints each digit of a number in reverse order.
4. Demonstrate usage of type casting with multiple conversions in one program (int → float → char).
5. Implement a conditional fine policy using if-else-if-else ladder for various return cases in a digital kiosk.

Group G:

1. Design an algorithm and flowchart to calculate factorial of a number using recursion.
2. Write a C program to demonstrate advanced I/O using sprintf, sscanf for structured data processing.
3. Create a calculator program that computes area and perimeter of multiple shapes using user menu and function calls.
4. Input multiple logical conditions and evaluate their outcomes using logical tables and printf formatting.
5. Validate a triangle using input lengths and classify it further as right-angled, acute or obtuse using if-else.